SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID – 25130**
- **Problem Statement Title - Student Innovation**
- **Theme- Heritage & Culture**
- **PS Category- Software**
- **Team ID-**
- **Team Name- Netaji Ninjas**

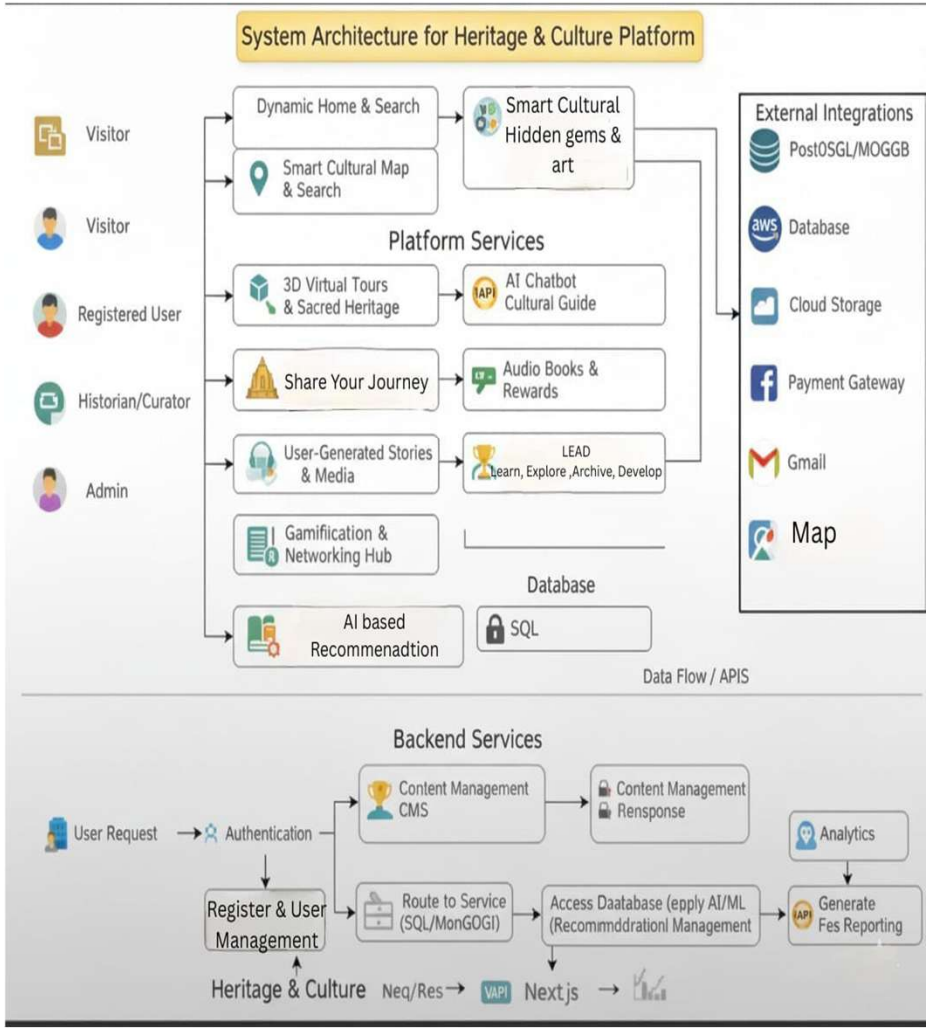
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BharatVerse

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Proposed Solution

Comprehensive Heritage & Culture Platform: One-stop hub to preserve, promote, and experience cultural heritage.
Dynamic Home Page: Explore States, Heritage Sites, Festivals, Art Forms, Hidden Gems easily.
AI Chatbot Cultural Guide: Instant assistance for heritage, culture, and travel queries.
3D Virtual Tours & Sacred Heritage Access: Immersive 360° experiences of monuments and festivals.
User-Generated Stories & Media: Share journeys, **local hidden places**, and images, verified at backend.
Audio Books & Hidden Gems: Narratives of ancient sites and unexplored cultural treasures.
Gamification & Rewards: Quizzes and challenges with points to boost engagement.
Smart Cultural Map & Search: Quick discovery of sites, events, traditions.
Smart Cultural Networking Hub: AI-driven recommendations of events, sites, art forms, and communities.
LEAD Program (Learn, Explore, Archive, Develop): Continuous learning, archiving, and cultural literacy.



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TECHNICAL APPROACH

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Frontend: React.js + Vite, React Native, Redux, Next.js, Tailwind CSS, Framer Motion, jQuery, JavaScript, CSS

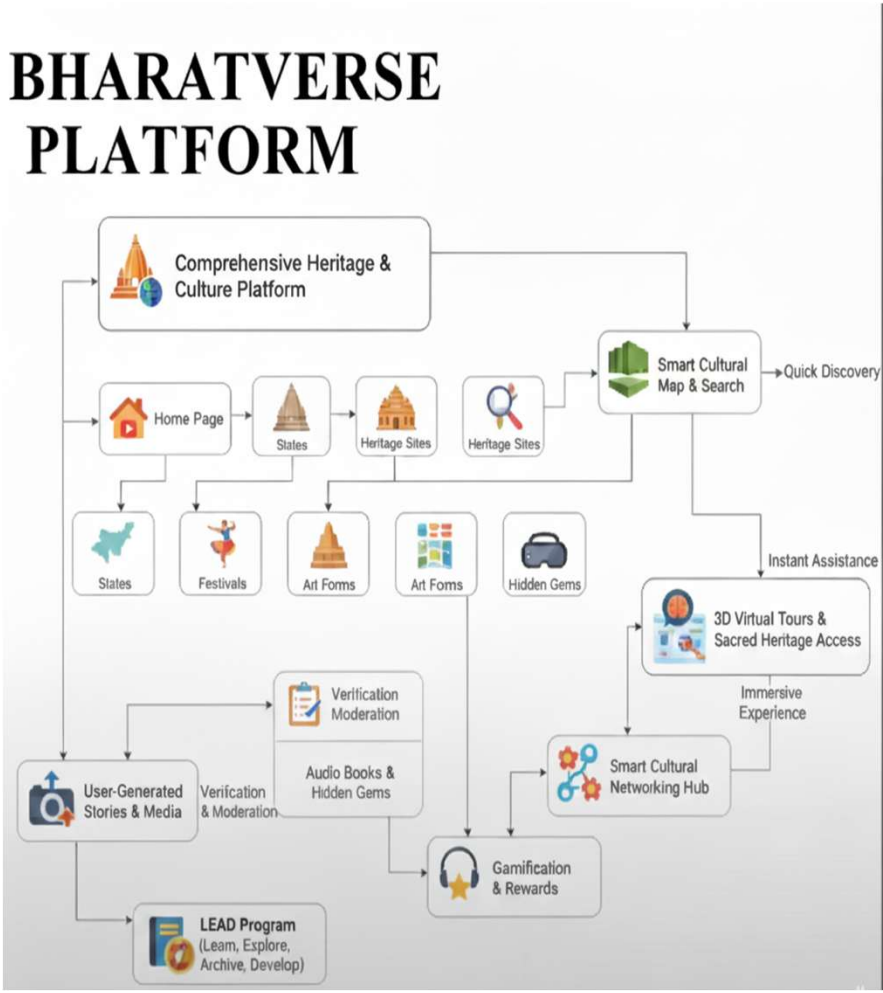
Backend: Node.js, Express.js, REST API, MongoDB, SQL, Supabase, Firebase, Socket.IO, Solidity

AI & ML: Chatbot (ChatGPT-4, Gemini API), TensorFlow / PyTorch (custom ML), AI-based recommendation

Services & APIs: MeiliSearch, LinkedIn API, GitHub API, Gmail API, Auth0, Razorpay, Twilio, Cloudinary, Location Service API

Multimedia & Features: Audio Books (TTS/STT), 3D Monument Viewer (Panorama, WebRTC), Map & Hidden Gems

Deployment & Tools: AWS, GCP, Docker, Railway, Vercel, Kubernetes (k8s), Nginx, GitHub, GitBook, Postman, Jest



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FEASIBILITY AND VIABILITY

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🌐 Feasibility

•Technical – Uses Web, Mobile, AI, AR/VR & Blockchain.
•Modularisation – Phased rollout (core → AI → 3D).
•Market – Indian heritage tourism worth **₹16+ lakh crore**.
•Economic – Cloud infra = Low initial investment, scalable

💡 Viability & Business Potential

•Large User Base – Students, tourists, diaspora, art lovers.
•Revenue Streams – Subscriptions, CSR, partnerships, marketplace.
•Scalable – From state-level to pan-India & global.
•Long-term Value – Heritage preservation & continuous engagement.

⚠️ Challenges & Risks

•Data privacy & security.
•Sustained user adoption.
•Rapid tech evolution (AR/VR).
•Authenticity of user-generated content.

- **1.7B+ domestic visits** in India (MoT, 2023).
- **Cultural tourism** growing at 15% CAGR (FICCI–EY, 2022).
- **42 UNESCO** World Heritage Sites in India.

🛠️ Strategies to Overcome

•Strong encryption & compliance.
•Gamification, quizzes & rewards.
•Continuous AI & AR/VR updates.
•Backend expert verification for stories & local gems.

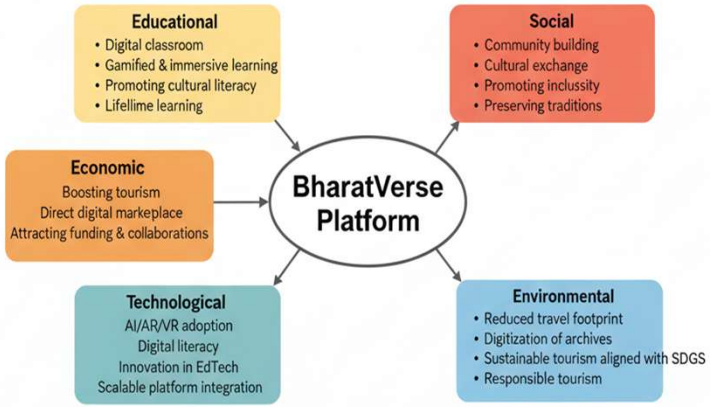
🏪 Business Model

1.Freemium Access – Free info + paid 3D tours, audiobooks, expert sessions.
2.Tourism Partnerships – **Incredible India, State Boards, ASI** (tickets & bookings).
3.CSR & Govt Support – Backed by **Digital India, Swadesh Darshan**.
4.Handicraft Marketplace – For artisans via **KVIC/Tribal India** (commission model).
5.Educational Subscriptions – Schools/colleges for gamified heritage learning.
6.Blockchain Donations – Transparent funding for heritage projects.

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IMPACT AND BENEFITS

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Target Audience

- **Tourists** – immersive journeys.
- **Students & Educators** – gamified cultural learning.
- **Artisans/Performers** – wider reach & verified sales.
- **Researchers/Historians** – authentic digital archives.
- **Govt/NGOs** – support cultural policy & preservation.
- **Diaspora** – deeper connection with Indian roots.

Social

- Builds pride in **42 UNESCO sites & diverse traditions**.
- Encourages sharing of **local hidden gems & stories**.
- Connects **urban & rural contributors**, preserving festivals & art forms.

Impact: +30–40% cultural literacy & participation.

Economic

- Boosts **domestic tourism (₹16+ lakh cr)** & local economies.
- Direct **digital marketplace for artisans & performers**.
- Generates jobs in **tourism, AR/VR, content creation**.

Impact: +25–30% artisan income, +20–25% tourism engagement.

Educational

- Functions as **digital classroom** with AR/VR, audiobooks & quizzes.
- Promotes **cultural literacy** in schools & lifelong learning globally.

Impact: +35–40% student engagement, +50% youth awareness.

Environmental

- **3D virtual tours** reduce travel & protect fragile sites.
- Digital archives cut **paper waste** & promote eco-tourism.

Impact: -15–20% carbon emissions, -40% paper usag

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RESEARCH AND REFERENCES

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Tourism & Heritage Platforms:

Incredible India: https://www.incredibleindia.org/
UNESCO World Heritage Centre (India): https://whc.unesco.org/en/statesparties/in
ASI (Archaeological Survey of India): https://asi.nic.in/
Google Arts & Culture: https://artsandculture.google.com/

Research & Best Practices

India Tourism Statistics 2023 (Ministry of Tourism, GoI): PDF
Analysing Visitor Trends at ASI Monuments (MoT, 2023): Report
Intangible Cultural Heritage (UNESCO India): https://ich.unesco.org/en/state/india-IN?info=elements-on-the-lists
Improving Heritage Management in India (NITI Aayog, 2023): PDF

Networking & Engagement Models

Tripoto Community Travel: https://www.tripoto.com/
Live History India: https://www.livehistoryindia.com/
FICCI Tourism sector page: https://ficci.in/sector/tourism-culture

Feature	Existing Platform	BharatVerse
Comprehensive Heritage & Culture Platform	✓	✓
3D Virtual Tours & Sacred Heritage Access	x	✓
User-Generated Stories & Media	x	✓
Audio Books & Hidden Gems	x	✓
Gamification & Rewards	x	✓
Smart Cultural Map Networking Hub	✓	✓
LEAD Program (Learn, Explore, Achieve, Develop)	x	✓